Politecnico di Torino

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Communications Systems

Assignment 1 Report

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In this assignment, we focus on code properties and performance and error detection.

Exercise 1

First, we generate two generator matrices and all information vectors. We know that codebook is defined as:

$$C = \{c = vG \ , V \in \ H^k\}$$

So, we generate all codewords and form the codebook. Then for each codeword Hamming weight, which is the number of bits equal to 1, is calculated.

In next step, we compute A_i (Multiplicity value) and A_i is defined as the number of codewords $c \in C$ with $W_H(c) = i \ , 0 \ \leq i \ \leq M$. Codewords, Hamming weights, and multiplicity values for $G1$ and $G2$ are presented in Table 1-1 and Table 1-2, respectively.

G1		G2	
Codeword	W_H	Codeword	W_H
00000000	0	00000000	0
10000111	4	10000110	3
01001011	4	01001011	4
11001100	4	11001101	5
00101101	4	00101101	4
10101010	4	10101011	5
01100110	4	01100110	4
11100001	4	11100000	3
00011110	4	00011110	4
10011001	4	10011000	3
01010101	4	01010101	4
11010010	4	11010011	5
00110011	4	00110011	4
10110100	4	10110101	5
01111000	4	01111000	4
11111111	8	11111110	7

Table 1-1: Codewords generated by $G1$ and $G2$ and their respective Hamming weights

A_i	C1	C2
A_0	1	1
A_1	0	0
A_2	0	0
A_3	0	3
A_4	14	7
A_5	0	4
A_6	0	0
A_7	0	1
A_8	1	0

Table 1-2: Respective multiplicity values for codewords in $C1$ and $C2$

Definition of Undetected Error:

The transmitter side sends C_T into Binary Symmetric Channel and at the receiver side, y is received with the equation $y = C_T + e$. At the receiver side, syndrome is computed through $s = yH$ in which H is parity matrix. Syndrome is zero if and only if $y \in C$ so only the codewords have syndrome equal to zero.

$$y = C_T + e \ , \ \ s = yH => s = (C_T + e)H = C_T H + eH \ , \ \ C_T H = 0 => s = eH$$

If e belongs to the C then $s = 0$ and y is accepted. But the problem arises when $e \in C \,, e \neq 0$. In this situation we suffer from an undetected error.

Probability of Undetected error is computed through this equation:

$$P(UE) = \sum_{e \in C \,, e \neq 0} P(e) \ , \ \ P(e) = P^{W_H(e)}.(1-P)^{M-W_H(e)} => P(UE) = \sum_{i=1}^{M} A^i.P^i.(1-P)^{M-i}$$

Using the multiplicity values obtained in the previous part and this analytical formula, we compute the probability of undetected error for given probabilities. The results are given in the Table 1-3.

Probability of error in BSC channel	P_{UE}(Code 1)	P_{UE}(Code 2)
10^{-1}	9.1854×10^{-4}	2.3×10^{-3}
10^{-2}	1.3448×10^{-7}	2.9206×10^{-6}
10^{-3}	1.3944×10^{-11}	2.9920×10^{-9}
10^{-4}	1.3994×10^{-15}	2.9922×10^{-12}
10^{-5}	1.3999×10^{-19}	2.9999×10^{-15}
10^{-6}	1.4×10^{-23}	3×10^{-18}

Table 1-3: Probability of undetected error for code 1 and code 2 using analytical formula

The plot for the analytical part is shown in Figure 1-1.

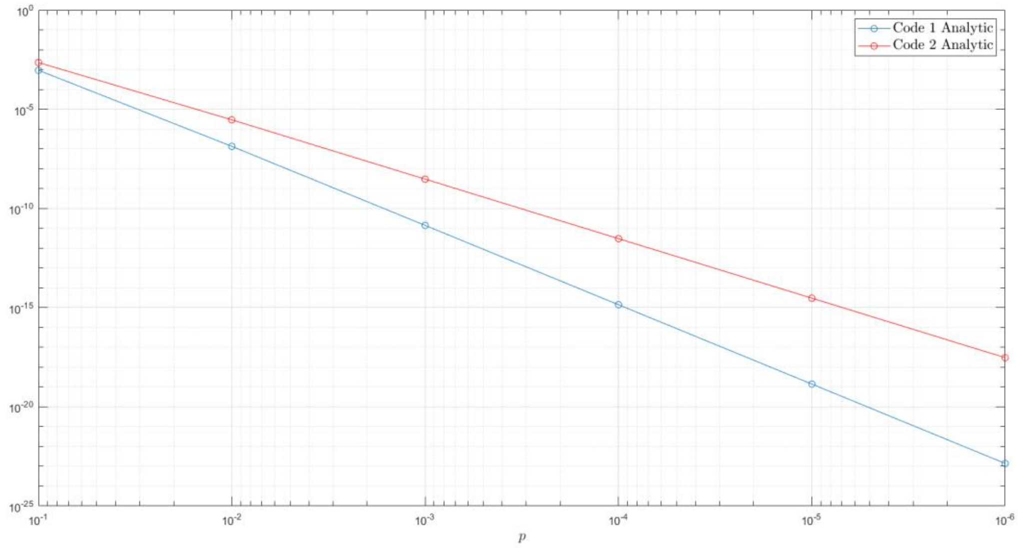

Figure 1-1: Displaying the probability of undetected error for code 1 and code 2 using analytical formula

For the second part of this exercise, at first, we generate $H1$ and $H2$. We know that parity check matrix H is made by a parity matrix p which exists in the generator matrix and an identity matrix with the diagonal size of r. In the next step, we generate a binary information vector v and we produce the transmitted codeword C_T by encoding the generator matrix G with v. Then we send the C_T over a Binary Symmetric Channel and obtain y. We compute the syndrome using the equation $s = yH$. As mentioned earlier, there is an undetected error if syndrome is zero and error vector belongs to the codebook. Equivalently, there is an undetected error if syndrome is zero and C_T and y are not same. We apply this rule to obtain required results for the simulation.

For doing simulation, we write a while loop and do the entire above-mentioned procedure inside it. The while loop is iterated until we get the desired number of undetected errors. Then probability of undetected error is number of iterations divided by number of errors. The results for code 1 and code 2 are given in Table1-4.

Probability of error in BSC channel	P_{UE}(Code 1)	P_{UE}(Code 2)
0.1	8.6583×10^{-4}	2×10^{-3}
0.09	6.1201×10^{-4}	1.8×10^{-3}
0.08	3.6147×10^{-4}	1.3×10^{-3}
0.07	2.5521×10^{-4}	10^{-3}
0.06	1.3498×10^{-4}	4.9459×10^{-4}

Table 1-4: Probability of undetected error for code 1 and code 2 using simulation

We plot all the results obtained from simulation and analytical calculations which is shown in Figure 1-2.

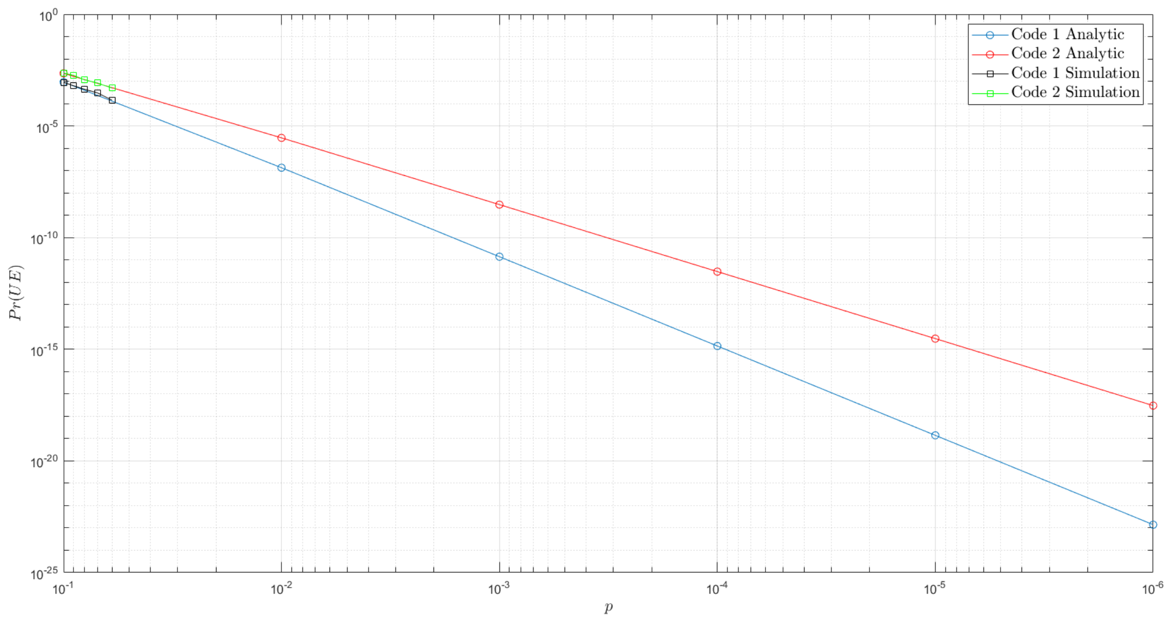

Figure 1-2: Displaying the probability of undetected error for code 1 and code 2 using both analytical formula and simulation

We observe that results obtained from analytical formula and simulation completely match each other. Also, we note that the difference between two codes performances lies in their distance profile and more specifically, their minimum distance. We know that having a large minimum distance enhances the performance for both error detection and error correction and design rule for error detection is to design a code with large d_{min}. First code has a minimum distance of 4 and the second one has a minimum distance of 3 and this justifies that code 1 has a better performance.

Exercise 2

We consider primitive polynomial $p(D) = D^3 + D + 1$ and multiply it by $D + 1$ to obtain generator polynomial $g(D) = p(D).(D + 1) = D^4 + D^3 + D^2 + 1$.

We know that $c(D) = v(D).g(D)$ and therefore we can encode the entire H^k which is set of all binary information vectors by using generator polynomial. For encoding we use the commands written in the assignment and we compute the distance profile for different values of k, ranging from 2 to 8.

We define the distance profile of a code as the set of all distances we obtain by fixing a codeword c and computing the $d_H(c, c')\forall c' \in C$.

We observe that all of distances in distance profile are even for all k. This is because $D = 1$ is a root of $g(D)$ and therefore is a root of $c(D)$ so all the codewords have an even weight and minimum distance cannot be an odd number.

An important observation is that for $k = 2$ and $k = 3$, minimum distance is 4 but for $k \geq 4$, minimum distance is 2. Now we want to discuss why this happens.

1) Since $g(D)$ has degree 4, when it encodes information vector v, the obtained codeword $c(D)$ has degree 4 more than $v(D)$ because $c(D) = v(D).g(D)$. For example, if $k = 2$, $v(D)$ has a degree of 1 and $c(D)$ has a degree of 5.
2) Also, we know that if $p(D)$ has a degree X, it does not divide any $D^y + 1$ for $y < 2^x - 1$. If we use $p(D)$ as a factor, all polynomials $D^y + 1$ cannot be codewords. Here in this exercise $p(D)$ has degree 3 and by applying the rule mentioned, all polynomials $D^y + 1$, $y < 7$ cannot be codewords.

For $k = 2$, y is 5 and for $k = 3$, y is 6 and when k has these values, $D^y + 1$ is not a codeword and we do not have a codeword with weight 2. But for $k \geq 4$ y is at least 7 and hence, $D^y + 1$ is a codeword and the codes with $k \geq 4$ have d_{min} =2 while for codes with $k = 2$ and $k = 3$, $d_{min} = 4$. Minimum distance for $k = 2$, $k = 3$ is equal to 4 because we showed that it cannot be equal to 2 and, all of the codewords have an even weight so minimum distance cannot have an odd value and therefore $d_{min} \neq 1{,}3$ for k=2,3 and hence it is equal to 4.

Exercise 3

For this question, at first, we must choose a primitive polynomial to detect errors with weight up to 3. To satisfy this condition, we look at the primitive polynomials properties. We know that if a primitive polynomial $p(D)$ has degree X then it does not divide any $D^y + 1$ for $y < 2^X - 1$ and when we use $p(D)$ as a factor, all polynomials $D^y + 1$, $y < 2^X - 1$ cannot be codewords.

Now we must find y to find a proper degree X for our primitive polynomial. When $k = 100$, $v(D)$ has a degree of 99 and $g(D)$ has a degree of $r = 8$ and therefore $c(D)$ has degree 107 which is the maximum degree (or in another word, worst case to satisfy). We must choose a X such that $D^{107} + 1$ is not a codeword to ensure that we do not have a codeword with weight 2. So, we have $107 < 2^X - 1 => \ \mathrm{X} > 7$. Therefore, we should choose a polynomial with degree 7 to ensure that we do not have a codeword with weight 2 and all error vectors with weight 2 cannot produce undetected error.

In order to detect errors with weight 1 and weight 3, we just have to multiply the primitive polynomial by (D+1). Since D=1 is a root of (D+1), it become a root of g(D) and therefore a root of c(D) and because of this, all of codewords have an even weight and errors with odd weights are detected.

From primitive polynomials table, we choose $D^7 + D^3 + 1$. We encode all 2^8 information vectors by using the given commands in the assignment and produce the codebook. Now, we compute the distance profile, minimum distance and multiplicity values.

We observe that minimum distance is equal to 4 and therefore error vectors with weight $W_H(e) < 4$ cannot produce undetected errors. Also we see that all values in distance profile are even and multiplicity values only exist for even values. This is because $D = 1$ is a root of $g(D)$ and therefore is a root of $c(D)$.

Now with multiplicity values A_i , we compute the probability of undetected error for given probabilities using analytical formula. The results are shown in Table 3-1.

Probability of error in BSC channel	P_{UE}
4×10^{-1}	3.8×10^{-3}
3×10^{-1}	3.4×10^{-3}
2×10^{-1}	2.2×10^{-3}
10^{-1}	4.7086×10^{-4}
5×10^{-2}	5.4535×10^{-5}
10^{-2}	1.4187×10^{-7}
10^{-3}	1.5809×10^{-11}
10^{-4}	1.5981×10^{-15}
10^{-5}	1.5998×10^{-19}

Table 3-1: Probability of undetected error using analytical formula

Now we use the same method as Exercise 1 to compute the probability of undetected error using simulation. Table 3-2 shows the results of the simulation.

Probability of error in BSC channel	P_{UE}
4×10^{-1}	3.8×10^{-3}
3×10^{-1}	4×10^{-3}
2×10^{-1}	2×10^{-3}
10^{-1}	5.1536×10^{-4}

Table 3-2: Probability of undetected error using simulation

We observe that the results of simulation and analytical formula are approximately same. Now we plot the obtained results which is shown in Figure 3-1.

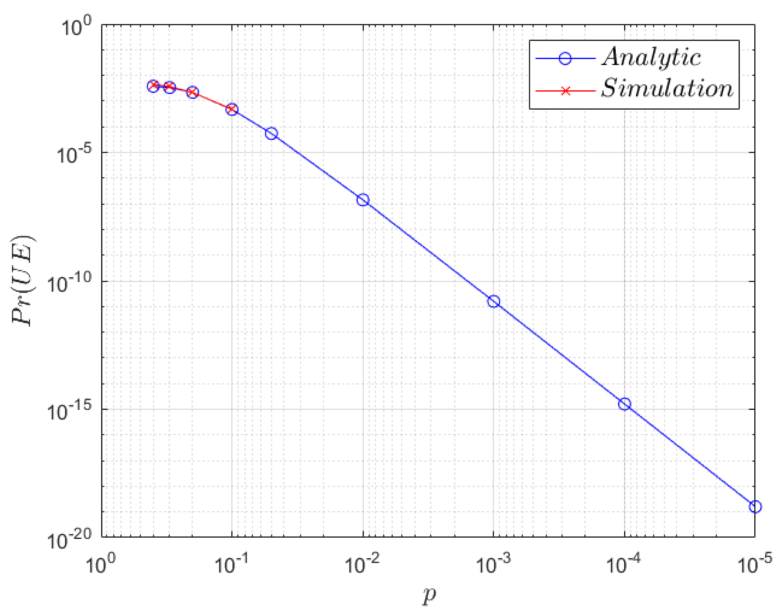

Figure 1-2: Displaying the probability of undetected error using both analytical formula and simulation

Question 1

As we defined earlier in Exercise 2, we define the distance profile of a code as the set of all distances we obtain by fixing a codeword c and computing the distance $d_H(c, c')\ \forall c' \in C$. Now, we want to show that, this distance profile is independent from reference codeword.

We know that $d_H(v_1, v_2) = W_H(v_1 + v_2)$. Also, we know that if we take C and we sum a codeword c to all codewords, we obtain the same codewords in another order.

Suppose we choose two codewords c_1 and c_2 from codebook C. Distance profile for c_1 is defined as: $d_H(c_1, c') = W_H(c_1 + c')\ \forall c' \in C$. By using above mentioned concepts, we know that $(c_1 + c')\ \forall c' \in C$ gives us the codebook in a different order from previous order so we have $d_H(c_1, c') = W_H(\hat{c}_1)\ ,\ \hat{c}_1 \in C$. Like c_1 , Distance profile for c_2 is defined as: $d_H(c_2, c') = W_H(c_2 + c')\ \forall c' \in C$, and $(c_2 + c')\ \forall c' \in C$ gives the codebook in another order and we have $d_H(c_2, c') = W_H(\hat{c}_2)\ ,\ \hat{c}_2 \in C$. So, we observe that computing the distance profile is independent from starting codeword and therefore, for computing the distance profile, we just need to compute Hamming weight for all codewords in codebook and we can say that distance profile is the set of the Hamming weights for all codewords in codebook.

Now we want to show this in an example. Suppose we have a code $C(4,3)$. We take $c_1 = 1100$ and $c_2 = 0101$. In the Table 4-1, we observe the Hamming weights of codewords and distance profiles with these two codewords as the reference.

Codeword	Hamming Weight	$c_1 + c$	$W_H(c_1 + c)$	$c_2 + c$	$W_H(c_2 + c)$
0000	0	1100	2	0101	2
0011	2	1111	4	0110	2
0101	2	1001	2	0000	0
0110	2	1010	2	0011	2
1001	2	0101	2	1100	2
1010	2	0110	2	1111	4
1100	2	0000	0	1001	2
1111	4	0011	4	1010	2

Table 4-1: Distance profile of C(4,3)

As it is shown, First, third, and fifth column have same values but in different orders as we said before, and therefore computing the distance profile is independent from starting codeword. Also, it is observed that distance profile is the set of the Hamming weights for all codewords in codebook.

Question 2

We show and write the expressions in Table 5-1.

	$N_{TX} = 1$	$N_{TX} = 2$	$N_{TX} = i$
$P(A;C)$	P_0	$P_0 + P_0P_2$	$P_0 + P_0P_2 + \cdots + P_0{P_2}^{i-1} = P_0 \sum_{j=0}^{i-1} {P_2}^j$
$P(A;UE)$	P_1	$P_1 + P_1P_2$	$P_1 + P_1P_2 + \cdots + P_1{P_2}^{i-1} = P_1 \sum_{j=0}^{i-1} {P_2}^j$
$P(NA)$	P_2	P_2^2	${P_2}^i$

Table 5-1: Expressions for probabilities in retransmission scenario

Now we want to show that they sum to 1.

$$P(A;C) + P(A;UE) + P(NA) = P_0 \sum_{j=0}^{i-1} {P_2}^j + P_1 \sum_{j=0}^{i-1} {P_2}^j + {P_2}^i$$

$$= P_0 + P_0P_2 + \cdots + P_0{P_2}^{i-1} + \ P_1 + P_1P_2 + \cdots + P_1{P_2}^{i-1} + {P_2}^i$$

$$= P_0\left(1 + P_2 + \cdots + {P_2}^{i-1}\right) + P_1\left(1 + P_2 + \cdots + {P_2}^{i-1}\right) + {P_2}^i$$

$$= (P_0 + P_1)\left(1 + P_2 + \cdots + {P_2}^{i-1}\right) + {P_2}^i, \qquad P_0 + P_1 + P_2 \ = 1 => P_0 + P_1 = 1 - P_2$$

$$=> P(A;C) + P(A;UE) + P(NA) = (1 - P_2)\left(1 + P_2 + \cdots + {P_2}^{i-1}\right) + {P_2}^i$$

$$= \left(1 + P_2 + \cdots + {P_2}^{i-1}\right) - P_2\left(1 + P_2 + \cdots + {P_2}^{i-1}\right) + {P_2}^i$$

$$= \left(1 + P_2 + \cdots + {P_2}^{i-1}\right) - \ \left(P_2 + {P_2}^2 + \cdots + {P_2}^i\right) + {P_2}^i$$

$$= 1 + P_2 + \cdots + {P_2}^{i-1} - \ P_2 - {P_2}^2 - \cdots - {P_2}^i + {P_2}^i = 1$$

All the terms cancel each other and only the first term (which is equal to 1) remains and therefore $P(A;C) + P(A;UE) + P(NA) = 1$

Question 3

First part:

We know that if $y \in C$ then syndrome $s = yH$ is equal to 0. In this case, if syndrome is zero then we have:

$s = 0 => y.H = 0$

y is a $1 \times M$ vector and H is a $M \times r$ matrix therefore r is a $1 \times r$ vector. If r=3 then

$s = yH$, $H = [col_1 col_2 col_3] => s = [y.col_1 \;\; y.col_2 \;\; y.col_3]$

If $s = 0 => [y.col_1 \;\; y.col_2 \;\; y.col_3] = 0 => y.col_1 = y.col_2 = y.col_3 = 0$

For the matrices H_1, H_2, H_3 we have:

$y.H_1 = [y.col2 \;\; y.col3 \;\; y.col1] = [000]$

$y.H_2 = [y.(col1 + col2)\; y.col2 \;\; y.col3] = [y.col1 + y.col2 \;\; y.col2 \;\; y.col3] = [000]$

$$y.H_3 = [y.(col1 + col2)\; y.(col1 + col2) \;\; y.col3]$$
$$= [y.col1 + y.col2 \;\; y.(col1 + col2) \;\; y.col3] = [000]$$

So, we observe that if $s = y.H = 0$ then permutation of the columns or replacing them with their linear combinations still produce parity check matrix of original code.

Second part:

Permutation of the rows does not produce parity check matrix because columns of H are involved in dot product with y (not rows) and if we change the rows then the dot product of y and H is not equal to zero if $y \in C$. We give an example in the following for both parts to clarify the results.

Example:

$G=$

$$\begin{matrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{matrix}$$

$H=$

$$\begin{matrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}$$

$y = [1 \;\; 0 \;\; 0 \;\; 1 \;\; 1 \;\; 0]$

$s = y.H = 000$

Now we introduce H_1 whose columns are a permutation of columns of H and H_2 whose rows are a permutation of rows of H.

$H_1=$

$$\begin{matrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{matrix}$$

$H_2=$

$$\begin{matrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{matrix}$$

Now we compute syndrome for these two matrices.

$s_1 = y.H_1 = [000]$, $s_2 = y.H_2 = [111]$ so permutation or combination of columns still produce parity matrix, which is not true for rows.